(c) What do you mean by ladder diagram? Draw 3 inputs AND, OR and NOT gate PLC ladder diagram. 7

(d) Write notes in brief on fixed Input/Output and modulator Input/Ouput. 7

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Unit-V

5. (a) Discuss the PLC counter functions. 2

(b) Define steps in planning a programm for large process ladder diagram construction with details. 7

(c) Write short notes on : 7

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(i) Holding Registers with example

(ii) PLC Timer function with example

(d) Draw and explain 3 inputs universal building block gates Relay equivalent and PLC logic equivalent circuits. 7

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B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(New Scheme)

(EEE Branch)

MODERN INSTRUMENTATION TECHNIQUES

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any* ***two*** *parts from parts (b), (c) and (d) of each question.*

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Unit-I

1. (a) Define ratio correction factor of Transformer. 2

(b) Describe the comparison method using watt meter for testing of potential transformer. 7

(c) Draw the equivalent circuit and phasor diagram of CT. Also derive the expression for phase angle errors. 7

(d) Write short notes on : 3½+3½=7

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(i) Errors in measurement

(ii) Reduction of errors of instrument transformers

Unit-II

2. (a) Define semiconductor photo diode. 2

(b) What do you mean by transducer? Describe the selection of transducer with characteristics. 7

(c) Give the principle of strain gauge for Gauge factor. Calculate gauge factor for a single strain gauge having resistance of 120 Ω is mounted on steel cantilever beam at a distance of 0·15 m from the free end. An unknown force F applied at free end produces a deflection of 12·7 mm of the free end. The change in gauge resistance is found to be 0·152 Ω the beam is 0·25 m long with width of 20 mm and depth of 3 mm. The Young's Modulus for steel is 200 GN/m^2. 7

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(d) Write short notes on : 3½+3½=7

(i) Hall Effect Transducer

(ii) AC Tachogenerators

Unit-III

3. (a) Give the purpose of sample and hold circuit in Instrumentation. 2

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(b) Explain with suitable circuit diagram, working and application of X-Y Recorder. 7

(c) What do you mean by DAS? Describe the multi-channel DAS with block diagram and description. 7

(d) Write short notes on : 3½+3½=7

(i) Galvinometer type Recorder

(ii) General description of DATA LOGGERS

Unit-IV

4. (a) What is Programmable logic controller?

(b) Explain in details the component and operation of PLC.

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